MUHAMMED ADEYEMI ADEDIGBA

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SUMMARY

I am an engineering professional with a passion for innovative software solutions that benefit the AECO industry, especially in relation to enabling engineering simulation at scale. I have a strong background in computation mechanics, finite element analysis, and structural engineering; and my formal engineering education culminates with a joint master's degree in advanced solid mechanics (coordinated by engineering schools and industrial partners across France, Greece, and Italy). In all I do, I steadfastly uphold values of self-discipline, integrity, diligence, and respect.

EDUCATION

Joint Master's Degree in Advanced Solid Mechanics **Sep 2022 – Aug 2024**
Specialisation: *Mechanics of Structures* Location: *Europe*

- **Schools:** École Centrale de Lille and Université de Lille, **France**; National Technical University of Athens, **Greece**; and Università della Calabria, **Italy** (*website:* **strains.eu**)
- **Grades: 16.41**/20 (**FR**); **9.2**/10 (**GR**); **29.50**/30 (**IT**); **18.00**/20 (**FR**) Ranked **1st** on study track (**Transcript**)

Bachelor of Engineering in Civil Engineering **Oct 2015 – Aug 2021**
Specialisation: *Structural Engineering* Location: *Nigeria*

- **School:** University of Ilorin (*website:* **unilorin.edu.ng**)
- **Grade: 4.90**/5.00 GPA (First Class | **Transcript**); Best Graduating Engineering Student (ranked **1st**/774)

WORK EXPERIENCE

Ansys (Part of Synopsys), *Lyon, France* **Oct 2024 – Present**
Research and Development Engineer

- Ownership of the full development lifecycle of Ansys actions, enabling composable CI/CD workflows that automate the development, testing, security hardening, documentation, and release of Ansys OSS.
- Maintaining simulation data processing libraries (*PyDPF-Core/Post*) through API contributions, feature enhancements, project documentation overhaul, containerization, and adoption of best practices.
- Authored PyAnsys Infrastructure as Code, leveraging Terraform for ensuring scalable, secure, and automated management of PyAnsys resources across Azure and GitHub.
- General PyAnsys ecosystem support, including (but not limited to) technical reviews of Ansys open-source repositories; contributions to and support of multiple PyAnsys libraries; and participation in major Python conferences to promote Ansys's commitment to open-source projects.
- Authored PyAnsys Logo Generator for standardizing the look and feel logos for PyAnsys libraries as well as automating their generation and distribution.

INNOSEA (ABL Group), *Nantes, France* **Mar 2024 – Aug 2024**
Offshore Structural Engineering Intern

- Developed a Python-based conceptual design software for offshore wind turbine jacket foundations, which interfaces with SACS through python APIs for finite element analysis and structural design.
- Conducted training for internal employees on using completed modules of the conceptual design software while continuously receiving their feedback for improvement opportunities.
- Gained more competence in using Ansys MAPDL and SACS for stress and fatigue life assessment of offshore structures across conceptual, pre-FEED, FEED, and detailed engineering design phases.
- Investigated the codebase of PREDIN (a legacy in-house software) for technical documentation preparation to foster proper understanding of legacy code within the team.

Trilogy, *Austin TX, USA (Remote)* **Sep 2021 – Jul 2022**
Software Support Engineer

- Utilized programming and computer networking skills, with deep product knowledge and excellent troubleshooting to triage customer problems and provide technical solutions across multiple products.

- Triaged end-user issues under 15 mins SLA and quality-checked triaged problems with attention to detail, cutting average response time by more than 65%.
- Demonstrated effective communication skills by engaging clients in meetings for root cause analysis of technical software problems, achieving more than 95% client satisfaction.
- Facilitated efficient collaboration by coordinating interactions between stakeholders (product managers, internal users, and external customers) and the Development Team.

Hash Analytics, *Remote* **Jun 2020 – Jul 2020**
Data Analyst

- Utilized scientific Python libraries (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn) for data scraping, processing, visualisation, and detailed data analysis.
- Developed a statistical machine learning AI model for predicting employee attrition at companies based on historical data using Scikit-learn.

Chevron, *Lagos, Nigeria* **Mar 2019 – Aug 2019**
Project and Structural Engineering Intern

- Gained understanding of using SACS for FE analysis and design of offshore jacket structures.
- Gathered data points used to enrich the database of an in-house project cost estimation software.
- Developed a project schedule reporting system which improved project progress reporting time.
- Provided project control support for various Project Teams, contributing to 100% cost recovery with the JV partner and receiving the 2019 Chevron Human Energy Award for the achievement.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C, MATLAB, Maple, LATEX,
- **Finite Element Analysis**: Ansys (Mechanical, MAPDL), Abaqus, Gmsh, SAP2000
- **Structural Design and CAD:** SACS, ProtaStructure, Midas Gen, AutoCAD, Onshape
- **Software tools:** Git, Github, Azure, Docker, Linux

RESEARCH EXPERIENCE

Development of a Conceptual Design Software for OWT Jackets – Master's Thesis **Mar 2024 – Aug 2024**
Conducted review of renewable energy trends and offshore wind foundation technologies; developed a Python-based jacket foundation pre-sizing software to serve as the jacket model generation module of the full conceptual design software; established structural analysis and design workflows in SACS considering ultimate and fatigue limit states; laid the groundwork for future integration of the jacket generation module with the SACS Analysis Engine by defining the requirements of Python modules for automated analyses launch, automatic member redesign, and custom post-processing of SACS analyses results.

Optimal Design of an Offshore Jacket Platform – Bachelor's Thesis **Oct 2019 – Jun 2021**
Conducted bibliographic research; acquired jacket platform design data typical of Nigerian oil fields; carried out size optimisation of structural members under gravity, wind, and hydrodynamic loads; produced improved design through 29% reduction in platform weight; performed results validation with those from literature.

Publication: M.A. Adedigba and A.A. Adedeji (2021). Optimal Design of an Offshore Jacket Platform, Journal of Science and Engineering Production — Unilorin Science and Engineering Periodicals (USEP-JSEP)

SELECTED AWARDS AND RECOGNITIONS

- Erasmus Mundus Scholarship Award (2021)
- Nigerian Society of Engineers (NSE) Award for Best Graduating Faculty of Engineering Student (2021)
- University of Ilorin Award for Best Graduating Faculty of Engineering Student (2021)
- American Society for Quality (ASQ) EMEA Competition 2nd runner-up award (2020)
- National Society of Black Engineers (NSBE) Award of Academic Excellence (2018)

LANGUAGE PROFICIENCIES

- **English:** Native/Bilingual
- **French:** Basic